

Artificial Intelligence, the ethical challenges that it presents and how Governments have and should approach its regulation?

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Abstract

In this project I will begin by understanding what AI is, and how it works. This will inform me of its capabilities, which dictate its practical applications within society. I shall then analyse some of the ethical concerns that Artificial Intelligence presents. From this technical and ethical understanding, I shall judge the societal need for regulation. Then I will break down two case studies of regulation: the UK's approach and the EU's AI act. Finally, I will conclude the future capabilities of AI and the appropriateness of both studied cases.

What is AI?

Before defining what AI is I believe it is first important to define what a computer is. It is arguable when the first computer was invented, some believe it was the general machine by Charles Babbage, others would say it was the abacus. I believe the first theorisation of a reprogrammable computer was the Turing machine by Alan Turing in 1936. A modified version of the original Turing machine was later built in 1943 by Alan Turing, I believe this to be the first example of reprogrammable calculator that can be used as a computer. Computers have since become far more advanced since the 1943 universal machine and now we have mobile phones that can perform millions of operations per second compared to the 15 operation per second that Alan Turing's 1943 universal machine could perform(BB)

Secondly, we need to define intelligence, the Cambridge definition of being intelligent is "the ability to learn or understand things easily"(V). As humans we measure intelligence using the IQ (Intelligence Quotient) system, which evaluates a person's ability to solve a variety of tasks and plots the persons intelligence on a normal distribution. For IQ 100 is the median score and the standard deviation being +/- 15 from 100 where 100+ is above average intelligence(Q). This system allows us to gauge someone's ability to reason and understand new concepts. For computers intelligence is viewed differently, instead of the broad spectrum of tasks that are used in the IQ system to measure human intelligence, a computer's intelligence is measured by finding how quickly they can solve tasks with known solutions (i.e. how quickly it can calculate the first 100 prime numbers). This can also be plotted on a normal distribution that allows technicians to quantify how quickly a computer can solve a set task compared to another computer's.

These two forms of intelligence are vastly different in their methods and applications; however, computer programmers can bring these two forms of intelligence closer together through good software. The definition of software is the instructions given to a computer. When software is improved the processing power of a computer can be better utilised in a way the user desires. This better utilisation will close the gap that divides human intelligence and computer's intelligence.

In addition, to assess the gap between humans and computer we must consider that, humans have the ability to differentiate between interacting with typical computer programs and actual humans. People are able to do this because of the lack of "intelligence" displayed by typical computer programs(F). As a software AI aims to mimic the way that beings seem Intelligent.

Since the invention of personal computers, the expression “my computer is playing up” has become increasingly common and it is a way of articulating “I can’t understand what the computer needs me to do.” AI aims to minimise and hopefully eradicate this problem between computers and people so that people in the future do not need to learn “how to use a computer” and instead the computer can learn how its operator wants to use it. By taking this step and implementing AI. Humans will be better integrated with computers so that they better aid our lives instead of hindering them.

What are the different categories of AI?

Presently, there are countless models of AI and numerous routes that can be chosen to program an AI. In this chapter I aim to outline broad categories of AI and differentiate categories from one another, so the capability of each category is clearly defined and can later be assessed.

Currently, the AI’s available fall into a category called “Narrow AI” this means that the set of tasks that the AI can successfully perform is small and application specific. The first kind of AI in this category would be low functionality AI. An example of this would be the chess bot Deep Blue, which beat a grand chess master Garry Kasparov. Deep blue was built By IBM in 1996 and is a form of reactive machine AI (U), meaning it was trained with data (previous chess games) and then it used that data along with statistical analysis to produce its own moves. Due to the limited training data AI’s based on this method lack the ability to adapt and change with the times so I class this form of AI as having low functionality.

The second AI in this category would be high functionality AI. An example of this would be chat GPT (Generative Pre-trained Transformer) which was first publicly released in November 2022. Chat GPT is a form of limited memory AI (meaning there is limited memory of the past i.e. it only has data from the date it was created) this means it uses past data and present data to inform its decision. As a result of this the AI has an ability to add to its own data to improve its decision process therefore, I class it as having high functionality. It’s important to note with large language models such as Chat GPT large teams of people are employed to scrutinise the AI’s own additions to its data to make sure it doesn’t go rogue i.e. teach itself a non-human readable language.

In the above paragraph’s, I have covered the categories of AI in the present day. However, there are constantly new AI’s being developed which will not fit into the above categories and instead fit into theoretical categories of AI. The two categories I am going to look at are AGI (artificial general intelligence) and super AI.

Within AGI, “Theory of Mind AI” is a theoretical model which, outlines a theoretical AI program that can understand the thoughts and emotions of other entities. This has similarities to limited memory AI, however, differs as it has the ability to infer new ways to tackle problems. This is unlike limited memory AI which reflects on past experience and applies the findings to the present situation. This difference is what sets Narrow AI and AGI apart. AI can solve only a narrow set of problems whereas AGI can proficiently teach itself to solve new problems in new context’s using the findings of past experiences in different contexts.

Finally, “Self-Aware AI” is a theoretical model of Super AI that no longer requires human input as it has gained its own wants, needs and beliefs. Super AI builds from AGI however unlike AGI it is able to understand its own thoughts and ideas as well as those around it. as a result over time, it will generate its own beliefs and emotions that it will act on.

Different paradigms of AI.

In computer science the word paradigm refers to the way a program works. In this context the paradigm of an AI approximately defines the method an AI takes to reach its decision. The first significant paradigm of AI would be ML (machine learning), which has its own sub-categories.

The first category of ML would be supervised learning which takes labelled data of inputs and outputs and creates formulae to link them. These formulae are used by the AI to respond to its situation and decide the most appropriate course of action to give it the best chance of success. Due to data having to be labelled for supervised learning to work, it is considered a form of narrow AI because it can only respond to situations it is highly familiar with.

The step up from supervised learning is unsupervised learning which uses unlabelled data and groups it to find patterns. This paradigm has a variety of applications due to its ability to find new connections with little human input. However supervised learning is still a form of narrow AI because its training data dictates the set of tasks it can successfully perform. Due to its unsupervised nature, it possesses the ability to expand its capabilities with the input of a human to identify if its new set of connections are useful.

The last form of ML is called reinforcement learning which is like unsupervised learning with the key difference of having a feedback loop. Reinforcement learning works by creating an “agent” which is an AI. The agent will then perform actions, and a system will then feed back to the agent numerous performance metrics that will tell it if its actions were successful or unsuccessful. This feedback is then analysed by the AI and the AI modifies its decision-making process. After multiple iterations of this process the AI improves its ability to perform the goal task. While reinforcement learning allows an agent to program its own training data it is still considered narrow AI due to its inability to determine if its action was successful or not in unfamiliar scenarios.

Deep learning is presently a field still in development, however, has great potential. Currently deep learning is being used in a variety of applications, despite its many mysteries. The deep learning paradigm refers to layering the above AI models on top of each other in a “neural network” to find even higher-level functions in data. These hard-to-find functions are very useful to businesses who constantly need to evolve and find competitive edges on their competition. Because of the complexity of deep learning algorithms, they are often referred to as “black box machines” due to their almost impossible to understand decision making process. Furthermore, due to the complexity of deep learning algorithms building them from scratch is a complex and expensive process.

Finally, foundation models are a paradigm of AI that partially resolves the two issues identified with deep learning models built from scratch. Foundation models are un-specialised deep learning models that are specialised by a programmer to perform a goal task. As foundation models are used by a variety of programs in a variety of applications, they are better understood. Programmers have collaborated to attempt mapping the decision-making process of the common foundation models. This helps them understand the model as well as helping new programmers understand how their program makes decisions. Furthermore, the cost of deep learning model construction is reduced when using a foundation model instead of building from scratch.

Current and future applications of AI

AI is currently being used in the banking industry particularly for fraud detection(B). The model often used for fraud detection is a form of machine learning most likely unsupervised learning. The data used to train these models is a register of past transactions and details such as if the transaction was fraudulent and the context behind each transaction. This application of AI is sensible because it is an augmented system where a human can intervene on the AI's decision. A human should be able to intervene with this process because the AI's decision is not always perfect and may need to be acted upon i.e. if the owner of the account specially requests a transaction go through.

The airline industry has used and is currently using a variety of AI's to price tickets for their flight(O). Initially airlines used reactive machine AI that statistically calculates the value of a ticket depending on factors such as: departure time, overheads for the airline, departure date and cookies attached to the customer's browser. The deployment of AI in the airline industry helps airlines make profit by appropriately pricing their often-low margin tickets. It is reported the average profit margin for tickets is around 1.2%. Meaning that a good AI can be the difference between a profitable business and a business doomed for bankruptcy.

It is a widely known fact amongst social media users that the content you interact with, is the content you will receive more of. This phenomenon is a direct result of an AI that monitors your interactions on the platform and makes evidence-based assumptions on content you will find engaging in future. LinkedIn (the social networking site) uses machine learning to: recommend jobs, suggest connections, offer online courses and rank suitable candidates for a job(A).

AI is also being deployed in the Automotive industry for a variety of applications such as collision avoidance and self-driving capabilities. The American nation highway traffic safety Administration reports that 93% of road accidents are caused by human error (impairment of judgment or senses due to factors such as: driver fatigue, a preoccupied mind and being under the influence)(Z). A study into the avoidance of collisions with the roll out of automated light vehicles (cars) predicted 56% of fatal crashes were avoidable(W). If automated vehicles were on the road compared to their human counterparts, this number of fatalities could further decrease with more road vehicles being automated and in communication with each other. While self-driving cars still have their issues the future for them is promising and will likely lead to a less dangerous road network. This result is expected due to a self-driving AI's ability to monitor more of its surroundings and pre-empt more dangers.

How can AI be unethical?

Britanica defines ethics as "the discipline concerned with what is morally good and bad and morally right and wrong"(Y). Ethics are a pillar of our society and without them society could run lawless. Therefore, when a new technology is being developed, it is vital that the ethics of the technology are studied so an informed decision about the technology's future place in society can be made.

In the past, humans have often not considered the wider effects of new technology and have instead immediately implemented them often without adequate regulations or tests to prevent harm to society. As a result, we now have past examples of technology poorly implemented that have caused widespread harm, such as the Oxycontin epidemic which caused immeasurable damage to families across the states. Due to the greed of Perdue Pharma and the negligence of FDA (food and drug administration) when it came to clinical trials.

Considering the facts above, I believe a comprehensive understanding of the ethics of, and surrounding AI is paramount to its optimal utilisation. The ethics of AI can be split into two categories: ethics for the developers and ethics for the consumers. Ethics for the consumers is the logical place to begin because companies need to consider their consumers ethics when deciding on their own.

Consumer concerns on privacy.

For many consumers privacy is the paramount concern with any digital service. In July 2015 a company by the name of “Ashley Madison” was breached by hackers and the personal data of its users was exposed(P). Ashley Madison was a dating site for married people and when the data of its users was released, millions of adulterers were exposed. While the ethics of a person being registered for Ashley Madison are arguable, it is arguable that the users of the site who entrusted the site with their personal data in confidence. should not of had their personal information exposed to the world.

I believe, as a consumer, it is vital to consider the data collection practice of any AI when it is handling potentially personal data. This is partly because the Ashly Madison case study (while extreme) demonstrates the disastrous repercussions of poor data handling practices.

It must be recognised that most AI’s will receive personal data as a result of their clients asking questions, to the AI containing personal data. Due to the black box nature of AI, I can deduce that it is vital that that personal data never makes it into the training data of the AI. The consequence of this happening would allow hackers the ability to scrape AI’s for personal data on specific people or allow them to take a personal data census’ of an AI’s users.

Consumer concerns on bias.

Relatively new to consumer concerns however, equally important is bias. People are becoming increasingly aware of bias in society particularly towards groups of marginalised people. From 2014 to 2015 Amazon was found to be using an AI program that had a bias against women during the recruitment process to corporate jobs(AA). While there are limited details about the AI, it was believed due to amazon using a cohort of primarily male programmers on the project. A skew in the training data occurred leading to a biased AI that would penalise CV’s containing words such as “women”.

Almost 10 years after the initial deployment of the AI by amazon there has been no legal action taken. This was partially due to the lack of the candidate’s knowledge of the AI’s existence and the employment decision not being “solely” made by the AI, which highlights the lack of legal framework surrounding AI. Furthermore, Amazon have since not officially commented highlighting the lack of social responsibility some companies hold.

In future it is vital that bias standards are implemented as a blanket across all AI’s and that the bias of an AI is regularly assessed. This would be to avoid incidents such as Amazons re-occurring and harming quality of life of groups of people with protected characteristics.

Consumer concerns on fake news.

Since 2022 and the release of chat GPT to the public. There has been a widely observed phenomenon of AI generated news articles. NewsGuard a company that tracks the reputability

of news sources found 49, news websites entirely AI generated. As a consumer it is becoming increasingly more difficult to tell fake news from what is real, especially for less technologically literate generations. This issue is ever growing and shall only accelerate with wider adoption of AI, presently there is little being done to manage and combat it.

Technologies such as fingerprinting and watermarking can be used to combat this issue, and such technologies would afford consumers the knowledge of if something is AI generated or not (provided the author is honest). These standards being implemented would likely stunt the growth of AI due to a society wide distrust in AI generated content. However, consumers will be better informed as a result and long-term trust in AI shall proportionally grow to how trustworthy it truly is.

Another solution would be for common software platforms such as windows to build in features for AI detection that users can opt into. These would be similar to technologies such as GPT zero however, they would require less effort to use, resulting in more people being informed about the AI generation chances of the content they interact with. Hopefully the knock-on effect of this would be, when people write they will be less informed by AI and more informed by other people. which will reduce the echo chamber effect that AI generated content has on other AI generated content. Hopefully stunting the effect that AI generated content has had on the spread of fake news.

Consumer concerns on access to harmful content.

Presently there are legal blanket safeguards in place to protect users of common internet platforms, from accessing content that is deemed harmful by moderators. Similar to common internet platforms, common AI's such as chat GPT-4 have safeguards in place which are effective against un-informed users. However, AI safeguards can still be bypassed and harmful content which is contained in the AI's training data can still be accessed(CC). There is little incentivisation for developers to make their harmful content avoidance better. This is because, there are no such safeguards in place for users of AI's. Instead, AI companies are expected to protect their users because it is morally right to do so however this responsibility is not shared industry wide.

Open AI, the company behind chat GPT took a firm stance on content moderation with both safeguards against harmful inputs and outputs. By taking these steps they aim to “empower businesses to maintain integrity, safety and moral standards”(X).

However, in 2023 Open AI was found to be employing low wage Kenyan workers who had the job of, marking sexually abusive language or hate speech in the AI's generated content (Ingram, 2023). This practice highlights the lack of blanket safeguards against the exposure of people to harmful digital content.

Developers concerns on over-regulation.

Goldman Sachs (A global investment bank) has predicted 7% growth of the global economy over the next 10 years as a direct result of the introduction of AI into global markets(M). This is massive potential and has the chance to uplift all of us into better standards of living, where more tasks are automated, and people can spend more time doing the things they want to do. However, some developers are concerned that regulation could “stifle innovation”(T) which would decrease the growth effect AI will have on the economy.

What issues do the Governments of the world need to address?

Considering the concerns from both consumers and developers of AI. Governments of the world must take appropriate measures to satisfy their citizens expectations(N). The Bletchley Declaration signed 1st of November 2023 by 39 individual nation representatives, was a step in the right direction for all 39 nations. The declaration states “AI should be designed, developed, deployed, and used, in a manner that is safe. In such a way to be human centric”. This declaration for many of the signing nations, is the guiding principle behind the steps that many of them will take to safeguard their populations from the dangers of AI.

To achieve this regulation Governments must carefully define AI and its sub-categories, so they are appropriate. They must then decide on regulations for each category of AI so that industry doesn't use AI's in ways that are unethical or already illegal. Then in cases where AI presents a risk that can be managed regulations must be written so that the AI can be used however there are appropriate safeguards in place to not endanger citizens.

What has the UK Government proposed?

The UK government in their national AI strategy aims to regulate AI in a way, “To remain an AI and science superpower, fit for the next decade” (Department for science, 2021). To achieve this goal, they plan to take an approach that is not sector-specific because the House of Lords believes “blanket AI-specific regulation, at this stage, would be inappropriate”(G). This approach was arrived upon because “regulators are best placed to consider the impact on their sector”(G). Some believe this approach is inappropriate due to the lack of protection it affords citizens that may be directly affected by decisions made by AI. However, others (such as I) align with this decision due to the leeway it gives innovators that aim to develop the technology, so it's potential is fully realised in ways not originally imagined.

Since 2018, when the House of Lords decided blanket AI regulations would be inappropriate, the UK Government has continued its journey of regulating AI, in a way that aligns with the goal of being “an AI and science superpower”. During this journey they have since identified that “The patchwork of legal frameworks that currently regulate some uses of AI may not sufficiently address the risks that AI can pose”(J). As a direct result of this assessment and in light of the fact that regulations can promote innovation, i.e. product safety legislation has increased innovation towards safer products(C). Which in the context of AI could constitute greater public trust in AI and increased adoption of the technology. The UK government has since begun work (such as publishing the AI white paper) to guide and regulate individual sectors on the adoption and use of AI.

The Department for Science, Innovation and Technology (DSIT), which recently absorbed the Department for Artificial Intelligence in February 2024, released guidelines for responsible AI in recruitment. These aim to promote good practice for the procurement and deployment of AI systems for HR departments i.e. not disqualifying a candidate solely based on the AI's decision. These guidelines are in line with the AI white paper and other key legislation such as The Equality Act of 2010. And they are based on the 5 principles laid out in the white paper which are:

1. Safety, security and robustness
2. Appropriate transparency and explainability

3. Fairness
4. Accountability and governance
5. Contestability and redress

The above five principles are seen being put into practice in the “responsible AI in recruitment” document. An example of this would be in section 3.2 titled “during procurement” the Department for Science, Innovation and Technology (DSIT) outlines that companies should not implement systems without requesting and later receiving relevant documentation such as “impact assessments, risk assessments” (R) which should be assured by valid methods such as bias audits (R). These guidelines assure the British public who will be receiving these AI informed decisions that they are fair which is in line with the five principles of responsible development of AI (J).

The UK Government has also begun taking steps to lead from the front. They have done this by publishing a guide to using Artificial Intelligence in the Public Sector (H). which encourages other government entities to adopt AI in a safe and ethical way to promote the implementation of AI across the economy. This advice informed the guidelines for the deployment of AI in recruitment and HR along with other approaches towards AI such as Ofsted’s approach towards the matter.

What has the EU Parliament proposed?

Presently the EU have agreed on a bill titled “The AI Act”, this bill is the first of its kind internationally and will come into force from 2024 to 2027 (K). The aim of these regulations according to the European parliament are to “ensure better conditions for the development and use of this innovative technology” (L). The bill is informed by their digital strategy that “will strengthen Europe’s capacities in new digital technologies and open new opportunities for businesses and consumers” (I). The EU hopes that the AI act will set a precedent that will encourage other nations to regulate AI in a similar manner so to avoid fundamental rights violations.

The AI act defines an “AI system” with a rather long definition, that can be summed up by “a system that infers how to achieve a given set of human-defined objectives using machine learning and/or logic” (L). This definition, in my opinion, is excellent, because it encapsulates all forms of AI in all applications which will allow the EU to properly regulate AI.

When drafting the AI act one of the greatest concerns about it becoming law was the potentially stifling effect it could have on innovation in the industry. Subsequently the EU has broken down all forms of AI into different 4 categories. Each category is defined by the risk it poses to people and in each category, steps are listed which must be taken to adequately address the risk the AI poses.

The categories are as follows: (K)

1. **prohibited AI systems;** these are systems that pose an extreme risk to individuals’ examples of prohibited systems would be: social scoring and categorisation, subliminal technique systems and systems that exploit disabilities (mental or physical). They are prohibited to negate the possibility of infringement on people’s personal liberties. There are a few exceptions that allow for the use of prohibited systems i.e. law enforcement may use biometric identification, however the EU must approve the systems’ use and an annual report must be submitted to the EU department of states for review.

2. **high risk AI systems:** these are systems that pose a risk to individuals if unregulated however may be used if the proper checks are in place. Examples of high-risk systems would be: any product on the EU market that previously had to undergo 3rd party conformity assessments, any systems listed in annex 3 of the act, any product that poses significant risk to the users health. high risk systems on the market must conform to transparency, documentation and cybersecurity requirements; by taking these steps products in this category will be safe for consumers.
3. **limited risk AI systems:** these are systems which don't necessarily pose a risk to the users health however the user should still be aware of the AI. Examples of systems in this category would be chatbots and deep fakes. Which will soon have to meet transparency requirements typically in the form of disclosure to the user that they are interacting with AI or AI generated content. Furthermore, for limited risk systems the user must always have the option to opt out of the interaction at any time i.e. chat with a human instead of a chatbot.
4. **unregulated AI systems:** these are systems which pose little to no risk to the user in any capacity. An example of these would be spam filters which needn't be regulated due to the minimal negative effect that they can have on people's lives.

Finally, to enforce these regulations fines have been announced for non-compliance. Placing a prohibited system on the market will lead to a fine of €35 million or a fine of 7% of annual turnover. Placing a high-risk system on the market will lead to a fine of €15 million or 3% of annual turnover. Placing a limited risk system on the market without being transparent will lead to a fine of €7.5 million or 1% of annual turnover(K).

Future categories of AI, their paradigm and how regulation should change

While it is impossible to know the future of AI, we can make predictions about where the industry will go in the near future and how it will get there.

As mentioned earlier deep learning is the most advanced paradigm presently and still is not fully understood. In the future we will see AI's become more advanced and more able to perform tasks that require typical human intelligence. I imagine much of this progress will be built from deep learning and further layering it. This may mean that we allow AI's of the narrow capability to interact with each other to solve laboratory problems. If successful we would eventually possess a "jack of all trades AI" that could turn its skills to any manner of tasks even new ones. Unfortunately, this would further contribute to the black box nature of AI. I do not believe this to be an issue, and this is because in many ways we hold AI's to the standard that we hold humans to. Following this judgement if the human brain is not fully understood however we trust it, then why don't we trust an AI that we don't fully understand.

This will present a challenge to regulators like no other and I have no doubt that regulations will need to be re-written according to the AI available at the time. Eventually, I imagine that AI law will no longer be contained in a single bill that dictates how humans may use AI. Instead, AI law will be contained in multiple different areas of Law which will each dictate acceptable AI practices within certain contexts. This approach is like the law that guides people and how they should act within society, and I believe with the eventual realisation of super AI this approach will be necessary for humans and AI to co-exist safely and harmoniously.

Conclusion

In conclusion there is no “right” approach to regulation of AI put simply we don’t yet know what, AI is capable of. However, not regulating isn’t an option. Currently AI poses limited risk to our society giving regulators a chance to properly consider their next step. I believe that the EU’s regulation of AI is fantastic due to their risk-based approach, and I hope that other nations follow by their example.

I envisage that soon individual sectors of the UK Government will begin their regulatory process in a similar fashion to the EU’s approach. I hope that we will have harsher punishments for not following regulations in high-risk sectors such as Law and Construction. This is because the current regulations that guide those industries are very strict, and I do not see why the same approach should be avoided just because AI is an emerging technology.

I also hope that we establish a national body that actively reviews each sectors approach over time so that regulations don’t fall behind AI in the future. The national body should also review the capabilities of AI’s so that unforeseen risk is properly mitigated especially in large machine learning models which are known to be unstable. Finally, subsidisation packages like the £50 million package announced by the ex-prime minister Rishi Sunak should be further funded through the guidance of this national body, so that UK remains’ a Science and Technology Powerhouse.

Evaluation

I appreciate that I have limited my scope to only the UK and Europe which is not a global picture. Other nations also face these challenges particularly for the technologically developed ones. However, judging from the information, I have been able to access, there has been less progress in their regulatory journeys when compared to the UK and Europe.

On balance, I recognise that it is difficult to predict the economic impact that current and future AI regulations will have on the market. Economic impact assessments are typically performed by highly skilled teams of individuals with rescores i.e. expensive modelling software’s. it would have been beneficial to perform this analysis myself however I chose to use the approximate but finished economic impact assessments from Goldman Sachs instead. using the finished assessments allowed me to better understand the effect of regulations on consumers instead. This may result in my conclusions on the regulations being too strict however through my judgement of the potential of AI I believe this to be a positive.

Issues that AI pose to society are ever growing and appearing. Subsequently, there are many that have only surfaced in recent months that I would have loved to have included. For instance, the recently studied surge in power demand from data centres that has been fuelled by the large-scale consumer adoption of AI. I imagine this issue will only further expand and presently I have seen no news of governments plans to tackle it or developers plans to minimise it.

I have found the EPQ to be hugely beneficial to my research skills. Particularly in my ability to access a sources validity by judging its reputation and evaluating its methodology. I have found it a challenge to articulate all my reading on this subject into my project. However, as a result of

this challenge, I have honed my research skills, especially when it comes to deciding if a source is relevant enough to read, understand and then later write about.

I found my project very interesting, understanding what AI is as well as understanding the process that Governments follow to regulate industries. In summary my fascination for this topic has only grown and in future I wish to study it further.

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